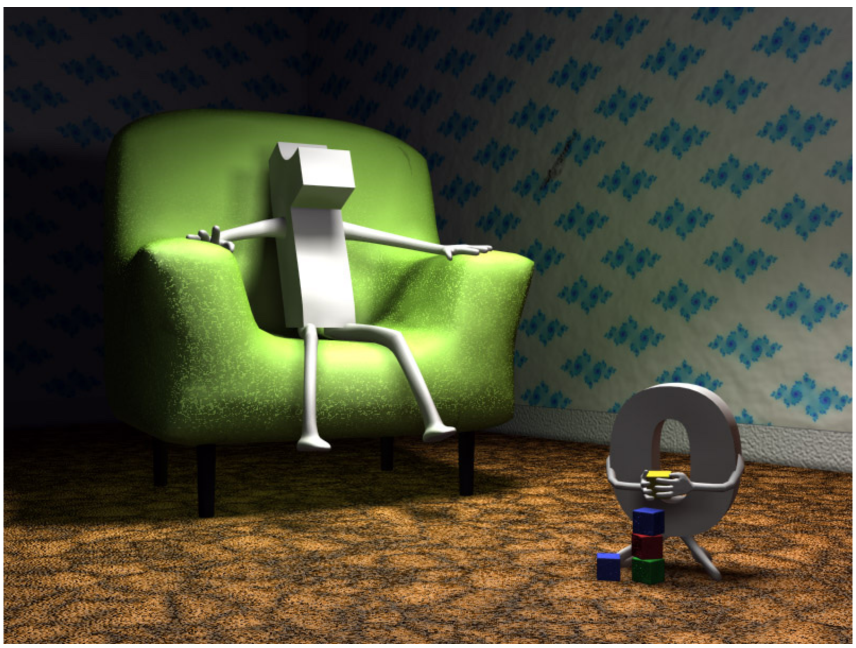
# Chapter 1. INTEGERS

## 1. 2 Divisibility and Prime Factorisation

### 1.2.1 Division Algorithm



**In this short video we prove the DIVISION ALGORITHM and see an example.**

**Division Algorithm.** Let $n$ and $d \geq 1$ be integers. Then there exist unique integers $q$ and $r$ such that $n = qd + r$ and $0 \leq r < d$.

quotient remainder

Proof. We divide the proof into two parts:

Part 1. Existence of $q$ and $r$.

Consider the set $X = \{ n - td \mid n - td \geq 0, t \in \mathbb{Z} \}$

Claim: $X \neq \emptyset$.

- If $n \geq 0$, then $n \in X$ because

$$n = n - 0 \cdot d \geq 0$$

- If $n < 0$, then $n - nd \in X$ because

$$n - nd = \underbrace{n}_{<0} \underbrace{(1-d)}_{\leq 0} \geq 0$$

Then $X \neq \emptyset$, and by the Well-Ordering Axiom we have that $X$ has a least element, say $r$. Then $\underline{r \geq 0}$ and $\underline{r = n - qd}$, for some $q \in \mathbb{Z}$.

So, $n = qd + r$. It remains to check that $r < d$. Suppose on the contrary that $r \geq d$. Then $r - d \geq 0$, and $r - d \in X$ because $r - d = n - qd - d = n - (q+1)d$. But this contradicts the fact that $r$ is the least element of $X$. So $\underline{r < d}$.

<u>Part 2</u>. Uniqueness of $q$ and $r$.

Suppose that we also have $q', r'$

such that $n = q'd + r'$, with

$0 \leq r' < d$.

We can assume without loss of generality that $r \geqslant r'$. Then

$$\left.\begin{array}{l} n = qd + r \\ n = q'd + r' \end{array}\right\} \Rightarrow 0 = (q - q')d + (r - r')$$

$$\Rightarrow (q' - q)d = r - r'$$

But $0 \leq \underline{r - r'} < r < \underline{d}$ and

$r - r' = (q' - q)d$, where $q' - q$ is an integer. From here, we get that

$q' = q$ and $r' = r$ ∎

Example. $n = -17, \quad d = 5$

$$-17 = \underbrace{(-4)}_{\text{quotient}} \cdot 5 + 3$$

remainder